

Project Proposal

UTHAO

End to End parcel delivery platform



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Problem Statement

Manual courier systems break down at scale.

- **Parcel loss**

Hard to locate items in transit

- **Incorrect delivery status**

Customers get wrong information

- **Hard to track movement**

No real-time visibility

- **Duplicate customer data**

Same person, many records

- **Poor warehouse management**

Stock badly organized

- **No performance reports**

Cannot measure agents

Proposed Solution

A centralized logistics platform **powered by PostgreSQL.**

- **Single Source of Truth**

No duplicate customer or parcel records

- **Automated Workflows**

Triggers & procedures run the logic

- **Real-Time Tracking**

Every parcel, every status update

- **Role-Based Access**

Customers, staff, agents & managers

Project Overview

- **Customers**

Create requests, track parcels, view history

- **Branch Employees**

Receive parcels, assign warehouses, update status

- **Delivery Agents**

Receive assignments, update delivery status

- **Managers**

Monitor operations, reports, performance

Main Features

- **User Management**
Registration & role-based access
- **Parcel Management**
Create, update, cancel, track
- **Warehouse Management**
Add, store, transfer parcels
- **Delivery Management**
Assign, schedule, confirm
- **Payment Management**
Charges, history, revenue tracking
- **Reporting**
Daily, monthly & agent performance

Role of RDB

- **Data Integrity**

Primary keys, foreign keys & constraints

- **Efficient Querying**

Joins, indexes & views for fast reports

- **Data Consistency**

Transactions keep operations in sync

- **Automation**

Triggers & procedures inside the database

Core Entities

- **Customer**
Sender & receiver info
- **Employee**
Branch staff
- **Parcel Assignment**
Agent ↔ parcel link
- **Branch**
Branch offices
- **Delivery Agent**
Delivery personnel
- **Status History**
Every parcel movement
- **Warehouse**
Storage locations
- **Parcel**
Details & current status
- **Payment**
Payment records

PostgreSQL Features

Course-relevant techniques we will demonstrate

- **Stored Procedures**

Assign & transfer parcels

- **Functions**

Delivery cost & tracking history

- **Triggers**

Auto status history & revenue

- **Views**

Active & delayed deliveries

- **Cursors**

Row-by-row monthly reports

- **Complex Joins**

Multi-table analytics

Tech Stack

- **PostgreSQL**

Database — procedures, triggers, joins

- **React**

Frontend — dynamic dashboards

- **Express.js**

Backend — REST API layer

- **Tailwind CSS**

Styling — clean responsive UI

Workflow

- 1** Customer books parcel
- 2** Branch receives parcel
- 3** Warehouse stores parcel
- 4** Manager assigns delivery agent
- 5** Agent delivers parcel
- 6** Customer confirms delivery
- 7** Payment completed

Expected Outcomes

- **Centralized parcel tracking**
One dashboard, all parcels
- **Automated reporting**
Generated on demand
- **Reduced data redundancy**
Normalized schema
- **Improved delivery monitoring**
Live agent status
- **Better warehouse management**
Clear stock control
- **Efficient revenue tracking**
Per-parcel billing

Advanced Features (Bonus)

- **Materialized Views**
Cached revenue summaries
- **Full-Text Search**
Search by name or tracking number
- **Partitioning**
History split by month / year
- **Transaction Management**
Atomic assign, rollback on fail
- **Audit Log Triggers**
Track every update & delete

Thank You

That's all for UTHAO!

